

Samuel Oluwakorede Oyefusi

[Email](#) · [Website](#) · [LinkedIn](#) · [GitHub](#)

RESEARCH INTERESTS

Robot Perception, Scene & Spatial Understanding, Autonomy for Mobility and Manipulation, Sensor Fusion.

EDUCATION

Oakland University — Michigan, USA

Ph.D., Electrical and Computer Engineering

Sept 2026 – Aug 2030

Advisor: Prof. Jared Strader

Worcester Polytechnic Institute (WPI) — Massachusetts, USA

M.S., Robotics Engineering

Jan 2025 – Dec 2026

GPA: 4.0 / 4.0

Pan-Atlantic University — Lagos, Nigeria

B.Eng., Mechanical Engineering (First Class)

Jan 2020 – Dec 2024

GPA: 4.79 / 5.0

2nd Overall Best Student, Faculty of Engineering

PUBLICATIONS & PREPRINTS

*(**Samuel Oyefusi** bolded in all author lists below)*

- [1] **Samuel Oyefusi**, Darlington Agholor, Lawrence Ukpoju, Peter Onu, “Development of a Cost-Effective Autonomous Ultraviolet-C Disinfection and Vacuum-Cleaning Robot,” *ICARA*, 2026.
- [2] **Samuel Oyefusi**, Oluwarantimi Bukola, Peter Onu, “The Potential of Generative Artificial Intelligence to Improve Learning in Schools,” *3rd International Conference on Sustainability: Developments & Innovations*, Springer Nature. *(Under publishing)*
- [3] Alexander Ozoani, **Samuel Oyefusi**, “Mortality of Distributed Understanding,” *Human Arenas*, Springer Nature, 2026, pp. 1–7.
- [4] **Samuel Oyefusi**, Raehan Kareem, Onoriode Avebenake, “Application of Vector Integral Theorems in Engineering,” Vol. 1 No. 2, 2026.
- [5] Samuel Obiagba, **Samuel Oyefusi**, Oluwaseun Martin, Peter Onu, Anup Pradhan “Easy PID Motor Controller (EPMC): A Cost-Effective Closed-Loop PID Motor Controller with Integrated Encoder Processing and ROS2 Support for Mobile Robotics,” *17th International Conference on Mechanical and Intelligent Manufacturing Technologies (ICMIMT)* .
- [6] Samuel Obiagba, **Samuel Oyefusi**, Peter Onu, ANup Pradhan “Easy Inertial Measurement Unit (EIMU): A Cost-Effective IMU Solution for Real-Time Orientation and Acceleration Estimation in Robotic State Estimation and ROS2 Systems,” 17th International Conference on Mechanical and Intelligent Manufacturing Technologies (ICMIMT)
- [7] Ifeoluwa Oduwaiye, **Samuel Oyefusi**, Ayoola Okunlola, Melody Alabi, “Utilization of Machine Learning Techniques in Predicting Burnout in Organizations,” *International Journal of Computer Applications*, Vol. 187, No. 113, 2026.
- [8] Abasiono Mbat, Oselumese Agbonrofo, Oladimeji Abaniwonnda, **Samuel Oyefusi**, "Code-Generated Tool Orchestration versus Native Function Calling in Stateful Business Workflows: A Multi-Model Benchmark Study" *International Journal of Computer Applications*, Vol. 187, No. 136, 2026.
- [9] Peter Onu, Anup Pradhan, Nelson Madonsela, **Samuel Oyefusi**, “Designing Resilient AI-Powered Digital Assistants for Industrial Applications: A Mini Review of Key Challenges,” *IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)* 2025.
- [10] Tito Osadebey, **Samuel Oyefusi**, Micah Udeogu, “Addressing Artificial Intelligence Bias through Inclusivity: A Case Study with Nigerian Food Images,” *2025 International Conference on Artificial Intelligence and Digital Ethics (ICAIDE)*, Guangzhou, China, 2025, pp. 469–473.

- [11] Mbanisi, Kenechukwu C. et al "State of Robotics in Africa"
- [12] Chima Okwuokei, Desmond Moru, **Samuel Oyefusi**"Virtual Reality Training System for Enhancing Skill Transferability in Volleyball Players" Preprints.org
- [13] Ambrose Osadugba, **Samuel Oyefusi**"A Comparative Floquet-Lyapunov Study of Perturbation-Induced Halo Orbit Instability in the Earth-Moon System." SSRN

PROFESSIONAL EXPERIENCE

TechnipFMC

Aug – Dec 2025

Project Engineering Intern

- Assisted in developing project plans, schedules, and resource allocations.
- Supported creation of technical documentation including drawings, specifications, and progress reports.

Nigeria LNG Limited

Jul – Sept 2024

Intern, Pipeline & Gas Supply

- Analysed 25 years of pigging operations data using advanced analytics tools, uncovering wear and corrosion trends that improved predictive maintenance planning.
- Developed an automated data logging system, improving data accuracy and real-time monitoring of pipeline conditions.

Orangewood

May – Jul 2024

Robotics Engineering Intern

- Developed advanced manipulation applications, enhancing robotic dexterity and efficiency in industrial environments.
- Contributed to Robo-GPT, refining automation functionality and user experience.
- Integrated MoveIt, ROS, and Gazebo simulation to improve cobot performance evaluation.

PROFESSIONAL SERVICES

Reviewer:

- International Journal of Social Robotics, Springer Nature (2026–present)
- International Conference on Industry of the Future and Smart Manufacturing (2025–present)
- The Artificial Life Conference, University of Waterloo (2026–present)
- ROSCon Diversity Scholarship Committee, Open Robotics (2026–present)
- Journal of Computer Science, Science Publications (2025–present)
- Journal of Mechatronics and Robotics, Science Publications (2025–present)

TEACHING

Black in Robotics

Summer 2025

Instructor — Introduction to ROS2 Summer Workshop

Elevate Them

2022 – 2025

Teaching Assistant

GET 501: Applied Design; GET 503: Engineering Simulation

Pan-Atlantic University

2022 – 2025

Undergraduate Teaching Assistant

- MEE 57: Industrial Automation & Control
- MCT 401: Introduction to Mechatronics
- MEE 501: Design of Robotics and Control Systems

Pan-Atlantic University

2022 – 2025

Peer Tutor

Core Mechanical Engineering courses

AWARDS & HONORS

- WPI Provost Scholarship, 2026
- Black in Robotics Travel Award, 2026
- Most Innovative Tech Award (Male) — Pan-Atlantic University, 2025
- 2nd Best Student Paper — 1st International Conference on Governance, Ethics & Technology, 2025
- 2nd Overall Best Student, Faculty of Engineering — Pan-Atlantic University, 2025
- Leadership Award Nominee — Pan-Atlantic University, 2025
- Mr. SST Award (School of Science & Technology spirit) — Pan-Atlantic University, 2025
- 2nd Place, African Mars Rover Competition — African Rover Challenge, 2024
- Innovator of the Year — Pan-Atlantic University, 2024
- Pan African Robotics Competition Finalist — PARC, 2024
- Young Innovators in Africa Nominee — African Innovation Conference, 2023

PROFESSIONAL MEMBERSHIPS

American Association for the Advancement of Science (AAAS) • National Society of Professional Engineers
• Project Management Institute (PMP®) • National Society of Black Engineers • Black in Robotics • Black in AI • National Postdoctoral Association

TECHNICAL SKILLS

Languages: Python, C++, MATLAB, R

Frameworks: ROS 2, PyTorch, OpenCV, YOLO, NumPy, Pandas, matplotlib

Simulation: Gazebo Harmonic, Isaac Sim, Omniverse, Ansys, Simscape

Design: Fusion 360, SolidWorks, AutoCAD, Inventor

DevOps: Git, Bitbucket, Docker

COMMUNITY & LEADERSHIP

Generation Unlimited

Jan 2025 – Jan 2026

Member, Young People Action Team

Developed and distributed skill-building resources to empower youth employability.

Pan-Atlantic University

Nov 2024 – Present

Alumni Class President

Organised industry excursions, secured fundraising, represented PAU at alumni events.

Pan-Atlantic University

Jul 2021 – Sept 2024

President, Living Green Club

Managed CSR collaborations, engaging corporate partners and organising workshops to promote climate action and sustainability on campus.

Theirworld

Ongoing

Global Ambassador

Advocate for SDG 4 (Quality Education) and SDG 7 (Clean Energy) at international youth forums.

SPEAKING ENGAGEMENTS & MEDIA

- Presenter — “Technology,” 1st International Conference on Governance, Ethics & Technology (GET1)
- Speaker — “Empowering Children and Young People to Lead Climate Action,” UNICEF World Children’s Day
- Panellist — “Fostering Youth-led Innovation for Agrifood System Transformation,” UN World Food Forum
- Speaker — “The Future of Africa’s Space Industry: Capacity Building for Sustainable Development,” [SGAC](#)

- Panellist — “Grad School Series: The Application Process,” Machine Learning Lagos Community
- **Media:** Neural Narratives ([#ResearchDiary2](#)) • [BusinessDay](#) • [Al Jazeera](#) (Africa AI) • [Channels TV](#) • [YouTube](#) / [Spotify](#)